WORKING TITLE

by

Alvaro Belmonte

A dissertation submitted to Johns Hopkins University
in conformity with the requirements for the degree of
Doctor of Philosophy

Baltimore, Maryland
June, 2020

© 2026 by Alvaro Belmonte

All rights reserved

Abstract

My thesis is about bla bla

Thesis Committee

Primary Readers

First Lastname (Primary Advisor)
Assistant Professor
Department of ChangeMe
Johns Hopkins Bloomberg School of Public Health

First Lastname
Assistant Professor
Department of ChangeMe
Johns Hopkins Bloomberg School of Public Health

First Lastname
Assistant Professor
Department of ChangeMe
Johns Hopkins Bloomberg School of Public Health

First Lastname
Associate Professor
Affiliation1, and
Department1 & Department2 at
Johns Hopkins Bloomberg School of Public Health

Alternate Readers

First Lastname
Professor
Department of ChangeMe
Johns Hopkins Bloomberg School of Public Health

First Lastname
Assistant Professor
Department of ChangeMe
Johns Hopkins Bloomberg School of Public Health

Acknowledgments

Thanks!

Table of Contents

List of Tables

List of Figures

Chapter 1

Outline

1. Background/Preliminaries
   (a) (Maybe) Virtual double categories
   (b) Multicategories/operads
      i. Definition and examples
      ii. Operads difference
   (c) ∞-cosmos
      i. Defninition and example using simplicial sets
      ii.
   (d) Dendroidal Sets
2. Main Contents
   (a) (Maybe) some of the work on lax monoidal structures, but using the virtual double categorical setting.
   (b) Showing the model of infinity operads follows most of the axioms of an ∞-cosmos.

(c) Introduce the definition of weak $\infty-$cosmos and show similar results from preliminaries still follows

Chapter 2

Background

2.1 Multicategories

EXPLAIN THE SOURCE OF THIS MATERIAL. "Multicategories were first defined in ..." "Our exposition follows ..." yes Multicategories are a generalization of categories. In categories, arrows have a single source/domain to a single target/codoamin, in a multicategories the arrows have a finite sequence of sources/inputs to a single target/output.

Definition 2.1.1. A *multicategory* $\mathcal{M}$ consist of a set S of objects and for each $n \geq 0$ and each sequence $s_1, \ldots, s_n, t$ of objects a set

$$\mathcal{M}(s_1, \ldots, s_n; t)$$

of multiarrows/operations, thought of as taking n inputs of objects $s_1, \ldots, s_n$ respectively to the object output t. Moreover, there are three kinds of structure maps:

- For each object s, an identity/unit $1_s \in \mathcal{M}(s; s)$

- For any sequence of objects $s_1, \dots, s_n, t$ and any n-tuple of sequences $d^i_1, \dots, d^i_{k_i}$ of objects for $i = 1, \dots, n$, a composition function of multiarrows/operations

$$\gamma : \mathcal{M}(s_1, \dots, s_n; t) \times \prod_{i=1}^{n} \mathcal{M}(d^i_1, \dots, d^i_{k_i}; s_i) \to \mathcal{M}(d^1_1, \dots, d^1_{k_1}, d^2_1, \dots, d^n_{k_n}; t),$$

usually written $\gamma(p, q_1, \dots, q_n) = p \circ (q_1, \dots, q_n)$ or even just $p(q_1, \dots, q_n)$

Furthermore these structure maps have to satisfy a number of axioms, which express that composition is unital and associative (Heuts and Moerdijk, 2022)

Remark 2.1.2. Let $\mathcal{M}$ be a multicategory, $f \in \mathcal{M}(s_1, \dots, s_n; t)$ and $g \in \mathcal{M}(d_1, \dots, d_l; s_i)$ for some $1 \leq i \leq n$. It will be convenient to have the following notation:

$$f \circ_i g := f(1_{s_1}, \dots, g, \dots, 1_{s_n})$$

Thus, $f \circ_i g$ is the composition of f with g in the ith variable.

Example 2.1.3. You can think of the multiarrows in $\mathcal{M}(s_1, \dots, s_n; t)$ as n-ary operations for a fixed set X as follows. Let M be the multicategory with a single object X, then the multiarrows in $M(X_1, \dots, X_n; X)$, where all X_i are the same object X, is the set of functions

$$f : X \times \cdots \times X \to X$$

where the domain is the n-fold cartesian product of X with itself.

Example 2.1.4. We can up the level from the previous example, by fixing a family of sets $\{X_c\}_{c \in C}$ indexed by a set of objects C. Then you can think of the multiarrows in $\mathcal{M}(c_1, \dots, c_n; c)$ as n-ary operations, taking inputs of types $c_1, \dots, c_n$ respectively to an output of type c.

For example, for a fixed family of sets , define a multicategory $\mathcal{M}$ whose objects is the set of colors C, and whose multiarrows are defined by $\mathcal{M}(c_1, \ldots, c_n; c)$ to be the set of functions

$$p : X_{c_1} \times \cdots \times X_{c_n} \to X_c$$

Composition in $\mathcal{M}$ is defined as composition of functions.

Example 2.1.5. The previous two examples are specific instances for this more general example.If $\mathcal{C}$ is a symmetric monoidal category and C is a set of objects of $\mathcal{C}$, then one obtains a multicategory with this set of objects by defining $\mathcal{M}(c_1, \ldots, c_m; t)$ to be the set of morphisms $c_1 \otimes \cdots \otimes c_n \to t$ in $\mathcal{C}$. We will sometimes write $\mathcal{C}^{\otimes}$ for the multicategory obtained in this way, when C is the set of all objects of $\mathcal{C}$ (granted it is small).

Example 2.1.6. A (small) category $\mathcal{C}$ can be considered as a multicategory $\mathcal{C}_m$ which only have unary arrows. I.e. the set $\mathcal{C}_m(c_1, \ldots, c_n; c)$ is always empty unless $n = 1$.

We can observe that multicategories from a category $\mathcal{MC}\mathsf{at}$.

Definition 2.1.7. For two multicategories $\mathcal{M}$ and $\mathcal{N}$, a *multifunctor* $\phi : \mathcal{M} \to \mathcal{N}$ consist of:

- A function $f : O_{\mathcal{M}} \to O_{\mathcal{N}}$ between the corresponding sets of objects
- For each sequence $s_1, \ldots, s_n, t$ of objects of $\mathcal{M}$, a map

$$\phi_{s_1,\ldots,s_n,t} : \mathcal{M}(s_1, \ldots, s_n; t) \to \mathcal{N}(f(s_1), \ldots, f(s_n); f(t)).$$

These maps being compatible with compositions and units in the obvious way.

2.1.1 Trees

Definition 2.1.8. A *tree* T consist of a finite set V of *vertices*, a nonempty finite set E of *edges*, a distinguished element $r \in E$ called the *root*, together with the following data:

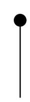
(1) a function $I : E - \{r\} \to V$, which we think of as assigning to an edge e the vertex $I(e)$ of which is an input.

(2) A function $O : V \to E$, assigning to each vertex v its output edge $O(v)$

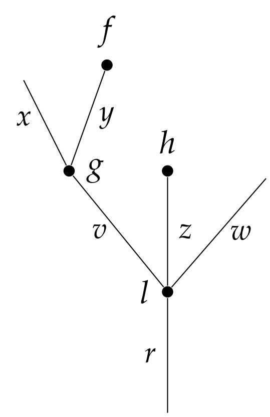
For each edge e other than the root r, we obtain a sequence of edges starting at e by repeatedly applying $O \circ I$. We demand that this sequence ends in the root r after finitely many steps, for an arbitrary starting edge. The edges in the complement of the image of O are called *leaves* of the tree T. The vertices in the complement of the image of I are called *stumps*, or *nullary vertices.* An *outer edge* is an edge tha is either the root or a leaf. An *inner edge* is any other edge of T. Such an edge is both an output edge and an input edge for some other vertex. The vertices come in two kinds, *external vertices* which only have one inner edge attached to them, and *internal vertices* When writing T for a tree, we will usually write $E(T)$ and $V(T)$ for its sets of edges and vertices, repectively.

Example 2.1.9. The smallest possible tree consiste of a single edge and no vertices; this tree will be denoted by μ

Example 2.1.10. The next smallest tree consist of a single edge and single vertex, pictured as:



Example 2.1.11. A bigger and more typical tree T with $E(T) = \{x, y, z, w, v, r\}$ and $V(T) = \{f, g, h, l\}$:



We will always pictures trees with the root at the bottom and the leaves at the top. The valence of a vertex v is the number of input edges it has, i.e. the cardinality of the set $I^{-1}(v)$. A vertex of valence 0 is what we called a *nullary vertex* or a *stump*. The sets $E(T)$ and $V(T)$ both have a natural partial ordering, where $x < y$ for edges x and y (or $f < g$ for vertices f and g) if there is a sequence of edges starting at x (resp. f) and ending at the root of T, r, and passes through y (resp. g).

A tree T defines a multicategory $\Omega(T)$ as follows:

- The objects of $\Omega(T)$ are the edges of T.
- An operation/multiarrow in $\Omega(T)(e_1, \ldots, e_n; e)$ is a vertex v of T such that the input edges of v are exactly $e_1, \ldots, e_n$ and the output edge of v is e. Note that for any sequence of colors $e_1, \ldots, e_2, e$ there is at most one operation/multiarrow in $\Omega(T)(e_1, \ldots, e_n; e)$.
- Composition is given by the tree structure in the obvious way. In more detailed, is the operation/multiarrow given by squashing an inner edge and identifying the two vertices to the composition of the two corresponding operations/multiarrows.

2.1.2 The Tensor Product of Multicategories

Let M and N be multicategories. Their Boardman-Vogt tensor product $M \otimes_{BV} N$ is a multicategory defined to make sense of what a M-algebra in the category of N-algebras is and viceversa. We will not talk about the algebras of a multicategories here, but we refer the reader to We define $M \otimes_{BV} N$ in terms of generators and relations. The set of objects of this multicategory is the product of the sets of objects of M and N; if x and y are objects of M and N respectively, we write $x \otimes y$ for the corresponding object of $M \otimes_{BV} N$. The generating multiarrows of $M \otimes_{BV} N$ are of two types:

- For each multiarrow $f \in M(x_1, \dots, x_n, x)$ and object y of N, a multiarrow

$$f \otimes y \in (M \otimes_{BV} N)(x_1 \otimes y, \dots, x_n \otimes y; x \otimes y)$$

- For each object x of M and multiarrow $g \in N(y_1, \dots, y_m; y)$, a multiarrow

$$x \otimes g \in (M \otimes_{BV} N)(x \otimes y_1, \dots, x \otimes y_m; x \otimes y)$$

The relations to be satisfied by these generators are the following:

(1) $(f \otimes y)(f_1 \otimes y, \dots, f_n \otimes y) = f(f_1, \dots, f_n) \otimes y$

(2) $(x \otimes g)(x \otimes g_1, \dots, x \otimes g_m) = x \otimes g(g_1, \dots, g_m)$

(3) $\sigma^*(f \otimes y) = (\sigma^* f) \otimes y$ for $\sigma \in \Sigma_n$

(4) $\sigma^*(x \otimes g) = x \otimes (\sigma^* g)$ for $\sigma \in \Sigma_m$

(5) $(f \otimes y)(x_1 \otimes g, \dots, x_n \otimes g) = \sigma^*_{n,m}((x \otimes g)(f \otimes y_1, \dots, f \otimes y_m))$

Note that relations (1) and (2) say precisely the condition that for any object y of N there is a multifunction $- \otimes y : M \to M \otimes_{BV} N$. Similarly, for any object x of M there is a multifunction $x \otimes - : N \to M \otimes_{BV} N$, this time given by relations (2) and (4). Relation (5) is the most interesting; most often refered to it as the *Boardman-Vogt interchange relation*. The permutation $\sigma_{n,m} \in \Sigma_{nm}$ is the one that makes sense of formula (5), i.e. the element relating the sequences $(x_1 \otimes y_1, \dots, x_1 \otimes y_m, \dots, x_n \otimes y_1, \dots, x_n \otimes y_m)$ and $(x_1 \otimes d_1, \dots, x_n \otimes d_1, \dots, x_1 \otimes d_m, \dots, x_n \otimes d_m)$.

2.1.3 Dendroidal Sets

Dendroidal sets are a generalization of simplicial sets that are particularly well suited to study operads and multicategories. Just as simplicial sets are presheaves on the simplex category Δ, dendroidal sets are presheaves on the dendroidal category Ω, which has trees as objects and certain morphisms between them. These trees are the same ones we defined in Section 2.1.1. We note that removing an external vertex and all the outer edges attached to it yields a new tree. We will refer to this procedura as *pruning*.

We will be extensively using the natural partial ordering on the set of edges. This partial order on the edges can be used to define *incoming edges* and *outgoing edges* for any vertex f, which we will denote by $in(f)$ and $out(f)$ respectively. The number of incoming edges of a vertex f is called the *valence* of f and we denote it by $|f|$. A vertex of valence zero is a *nullary vertex* or a *stump*. We will remember our first example of a tree, referred to as μ (Example 2.1.9). This tree will play a special role in what follows. For now we will just say that is has an unique edge which is both a lead and a root.

A tree is called *open* if it has no nullary vertices; on the other side, a tree is called *closed* if it has no leaves. As a quick example, the tree μ is open but not closed, and generally a tree cannot be both open and closed. A tree is called *linear* if all its vertives have valence one. A linear tree with $n+1$ edges running from the leaf 0 to the root n will be denoted by $[n]$:

The reason for this notation is that the linear trees will correspond to the standard simplices in the dendroidal category Ω.

Bibliography

Heuts, Gijs and Ieke Moerdijk (2022). *Simplicial and Dendroidal Homotopy Theory*. Springer Cham.